

PDP Paper-to-Notebook Reader Guide

A navigation companion for The Paired-Delta Protocol: A Within-Call Reference Architecture for Observer-QRNG Research

Purpose. This guide lets a reader move from a statement in the current paper to the notebook analysis that produced, checked, or contextualized it. Notebook locations are identified by visible markdown headers rather than cell or line numbers.

What each entry provides

- Paper section and claim
- Notebook name and visible header
- What question the analysis addresses
- How the analysis was performed
- The result produced by the executed analysis
- Dataset, population, and inferential unit
- Whether the notebook is primary, sensitivity, or external/not supplied

Important scope distinction. Study 1 / Pilot 4 results are produced primarily by NB1 and NB2. The Interleaving notebook is a retrospective Pilot 4 reconstruction used for partitioning sensitivity and limitations. The Exp4-versus-Exp5 notebook is a secondary cross-dataset check. The current paper also reports a preregistered held-out synthetic diagnostic-recovery validation and a separate 80-block local-coupling dataset. The validation analysis is included in this repository under preregistration/ rather than among the four primary analysis notebooks; the 80-block dataset remains a separate source. This guide marks those source boundaries explicitly instead of attributing them to NB1 or NB2.

Major updates from the prior guide. The map now reflects the revised paper title and section numbering, the condition-specific same-position mutual-information result in §3.2.5, the preregistered 32-cell artifact-injection grid in §3.3.4, scheduled-audit missingness in §3.4, the revised relational-source boundary in §3.7.2/S2.1, the new held-out validation in §3.8, and an explicit §4.4 analysis-role map distinguishing the planned Study 2 primary confirmatory test from mandatory diagnostics and exploratory relational analyses.

Notebook index

Where the supplied analyses live

Short name	Notebook	Primary contents
NB1	Exp4_Notebook1_Zen_Paired_Control_Stream_Design_Valid ation.ipynb	Design validation, PCS calibration, Baseline relational behavior, same-position MI, audit reconstruction, artifact injections including the locked confirmatory grid, and estimator/calibration checks.
NB2	Exp4_Notebook2_Zen_Temporal_Structure_Analysis.ipynb	Hit-rate results, single-scale H_RS analyses, session-count/threshold diagnostics, shuffle and metric cross-checks, within-session sensitivity, and lagged/local relational analyses.
Interleaving	Exp4_Interleaving_PhysicalHalf_AI_and_StreamLength_Audit. ipynb	Retrospective reconstruction of contiguous versus locally balanced partitions, deterministic odd-even allocation, AI comparison, and stream-length variance scan.
Exp4 vs Exp5	Exp4_vs_Exp5Prescreen_Baseline_CrossDataset_Check.ipyn b	Secondary comparison of Pilot 4 Baseline behavior with a dated Exp5-prescreen snapshot. Not a Pilot 4 primary inferential notebook.

NB1 and NB2 each exist in two functionally identical variants: `_Zen_` (local files only) and `_Git_` (falls back to downloading missing frozen inputs from this repository's `data/` folder on GitHub). Either may be used interchangeably for reproduction.

How to find an analysis

1. Find the paper section in this guide. 2. Open the named notebook. 3. Search for the visible header exactly or by its distinctive phrase. 4. Read the markdown immediately above the code for the estimand and limitations. 5. Read the saved output immediately below the code for the executed result.

When more than one notebook is listed, the first is the primary source unless the Source role field says otherwise. A paper claim can legitimately have no source in the supplied four-notebook set; those cases are marked explicitly.

3.1 / Figure 3

PCS calibration and condition comparisons

Notebook and header	NB1 - PCS Stream Calibration; PCS Randomness: Formal Mean Tests and Effect Sizes; PCS Distribution Shape; PCS Temporal Integrity
What it does	Uses all retained Pilot 4 PCS blocks to check mean target-aligned hit rate, condition differences in PCS hit rate and single-scale H_RS, and selected temporal/run diagnostics.
How it was done	Calculates the overall PCS hit-rate mean, session-clustered uncertainty, and sign-flip test against .50. Condition comparisons use session-level summaries/permutations; additional temporal checks preserve whole sessions.
Result shown	Overall PCS hit rate = 0.500253, 95% CI [0.499504, 0.501022], sign-flip p=.5160. No condition difference was detected for mean PCS hit rate (p=.7769) or mean PCS H_RS (p=.4696); the paper also reports no clear selected lag-1, run, drift, or dispersion abnormality.
Population / data	11,607 retained blocks; sessions preserved for inference
Source role	Primary result source

3.2.1

Baseline same-call Subject-PCS correlation

Notebook and header	NB1 - Analysis 1 - Same-call association diagnostic (under Baseline Empirical Paired-Delta Reference Distribution); Subject-PCS Association Diagnostics
What it does	Filters to Baseline and measures same-call association between simultaneous Subject and PCS scores, with session-aware uncertainty and within-session re-pairing/permutation.
How it was done	Computes Pearson and Spearman correlation for H_RS and target-aligned hit rate. Whole-session bootstrap supplies intervals; within-session permutations break original pairings while preserving session membership.
Result shown	H_RS Pearson $r=+0.0037$, 95% CI [-0.0306,+0.0371], permutation p=.8357; Spearman $\rho=+0.0027$, p=.8791. Hit-rate Pearson $r=+0.0149$, 95% CI [-0.0248,+0.0551], p=.3966. No tested same-call association resolved.
Population / data	3,090 Baseline blocks; 103 sessions
Source role	Primary result source

3.2.2-3.2.4

Baseline paired-difference behavior, physical halves, and re-pairing

Notebook and header	NB1 - Analysis 2 - Paired-delta location and shape; Empirical physical-half calibration for Paper §3.3.4 [header retained in notebook although current paper maps this result to §3.2.2]; Analysis 4 - Covariance contribution to paired-delta variance
What it does	Checks Baseline Subject-minus-PCS centering, physical-half A-versus-B symmetry before random labeling, and whether actual same-call alignment changes paired-difference variance relative to within-session cross-call re-pairing.
How it was done	Computes block-level deltas, resamples complete Baseline sessions, reconstructs physical halves from raw calls, and compares observed same-call variance/covariance with 10,000 within-session re-pairings.
Result shown	Baseline mean paired $H_{RS} = -0.0017$ with interval including zero; physical A-minus-B $H_{RS} = -0.00068$, 95% CI $[-0.00301, +0.00172]$. Observed paired-variance ratios were 0.9963 for H_{RS} and 0.9851 for hit rate, both inside their permutation reference intervals.
Population / data	3,090 Baseline calls; 103 sessions
Source role	Primary result source; notebook header contains an outdated paper-section pointer

3.2.5

Additional dependence and joint-stream diagnostics

Notebook and header	NB1 - Same-Position Bit-Level Coupling Diagnostics; Sampling-Hierarchy Check - Same-Position Mutual Information; Condition Comparisons of Subject-PCS Relationship Structure; Alternative Ordering Metrics Across All Conditions
What it does	Extends the Baseline checks across Human, AI, and Baseline using same-position mutual information, correlation/variance/joint-distribution comparisons, and a broad temporal/entropy battery.
How it was done	Same-position MI is calibrated within each block by circularly shifting PCS. Human uncertainty is contributor-clustered; Baseline/AI use session clustering. Separate condition contrasts and whole-session condition permutations avoid treating a significant-within-condition result as proof of a between-condition difference.
Result shown	Human same-position excess MI was negative and resolved: mean -0.000262 , 95% CI $[-0.000438, -0.000078]$, $p=.0058$. Baseline and AI were individually unresolved, and direct Human-Baseline ($p=.98$) and AI-Baseline ($p=.997$) contrasts were unresolved. Correlation/variance/energy comparisons were also unresolved; 47 of 48 extended temporal/entropy contrasts were unresolved and the remaining nominal result was compatible with chance in the unadjusted family.
Population / data	All Pilot 4 conditions; Human contributor hierarchy and automated-session structure preserved where implemented
Source role	Primary result source; this is a substantive update from the prior guide

3.3 / 3.3.1-3.3.3

Controlled artifact-injection operating behavior

Notebook and header	NB1 - Upstream Artifact-Injection Validation; Shared literal-bit bias; Shared persistence; Stream-specific control and unequal-half failure case
What it does	Starts from retained Baseline raw calls, applies specified disturbances before assignment/scoring, and measures what appears in Subject, PCS, and their paired difference. These are constructed operating-characteristic tests, not naturally occurring artifacts.
How it was done	Shared literal-bit and persistence disturbances are applied to both physical halves under synchronized and independent-position masks. Subject-only target replacement and unequal-half persistence provide cases where cancellation should fail. Whole sessions are resampled for uncertainty.
Result shown	Shared disturbances produced little systematic mean displacement of the paired difference while block-level variability increased. Subject-only replacement produced paired hit-rate changes about .0100, .0249, .0500, and .1000 at probabilities .02, .05, .10, and .20. Unequal-half persistence left a small systematic residual, showing the cancellation boundary.
Population / data	3,090 retained Baseline raw calls; repeated simulations
Source role	Primary result source

3.3.4

Preregistered locked artifact-injection grid

Notebook and header	NB1 - Preregistered Locked Confirmatory Injection Grid (locked 2026-08-19, run 2026-08-20/21); Both-metric (32-cell) breakdown of the locked confirmatory grid
What it does	Formalizes cancellation performance with a prespecified equivalence bound and fixed realization count after the exploratory injection work.
How it was done	Uses the same raw-call pipeline with $\delta=0.002$ and $R=100$ across 16 injection conditions, evaluated on both hit rate and H_{RS} for 32 statistic-specific cells. Four cells were disclosed as prior knowledge; the remaining 28 were prospective.
Result shown	30 of 32 cells had confidence intervals wholly inside ± 0.002 . The two persistence-dose .35 H_{RS} cells were indeterminate, not failures: independent-position $+0.00091 [-0.00052, +0.00222]$ and synchronized $+0.00113 [-0.00026, +0.00242]$. Tally: 30 PASS, 2 INDETERMINATE, 0 FAILS TO CANCEL.
Population / data	3,090 real Baseline calls reused across 100 simulated realizations per grid cell; 103 real Baseline sessions drive interval width
Source role	Primary preregistered result source

3.4 / Table 2

Audit availability, reconstruction, pass rates, and condition comparison

Notebook and header	NB1 - Scheduled-Audit Availability; Three-Subtest Audit-Battery Reconstruction and Session-Aware Summary; Audit-Failure Session Uniqueness - Stored-Only and Reconstructed
What it does	Reconstructs reached audit opportunities, quantifies missing audit records, reconstructs the implemented three-subtest battery from 1,000-bit sequences, verifies stored classifications, and compares failure rates by condition.
How it was done	Reached opportunities are inferred from frozen block progress. Frequency, Runs, and Longest Run of Ones are recomputed from audit sequences; 120 incomplete stored records are reconstructed; complete sessions are bootstrapped/permutated and condition comparisons Holm-adjusted.
Result shown	1,945 audit opportunities were reached; 1,891 audits were recorded, leaving 54 missing (Human 6, AI 47, Baseline 1). The reconstructed battery matched all 1,771 stored-complete classifications. Of 1,891 audits, 1,845 passed and 46 failed: 97.57% [96.89%,98.20%]. All Holm-adjusted condition-comparison p-values were 1.00. The paper cautions that availability differed by condition and provider/collection regime was confounded.
Population / data	1,891 recorded audits across 389 sessions; 1,945 reached opportunities across 390 sessions
Source role	Primary result source; updated to include scheduled-audit missingness

3.5 / Table 3

Mean-level target-aligned hit-rate results

Notebook and header	NB2 - Authoritative Four-Estimand Component Table; Population and AI Paired Hit-Rate Comparisons; Bayesian Hierarchical Hit-Rate Model
What it does	Reports Subject and PCS component means, within-call Subject-minus-PCS differences, Baseline contrasts, and the Bayesian companion model on the continuous paired difference.
How it was done	Frequentist uncertainty preserves Human contributors and sessions and preserves automated collection units as specified. The Bayesian model uses block-level Subject-minus-PCS as a continuous outcome with grouping intercepts and a Student-t likelihood.
Result shown	All Table 3 intervals for paired differences and Baseline contrasts included zero. Paired differences were +0.0201 percentage points Baseline, +0.0224 AI, -0.0200 Human-all, and +0.0873 Human 5+. The Bayesian model likewise estimated differences near zero: +0.018 pp Baseline, +0.019 AI, +0.060 Human 5+, with all 95% HDIs including zero.
Population / data	Baseline, AI, all Human, and descriptive Human 5+ subgroup
Source role	Primary result source

3.6.1
Full-Human single-scale H_RS result

Notebook and header	NB2 - Single-Scale Hurst - Population-Level Null
What it does	Compares the full Human paired H_RS mean/distribution with Baseline using hierarchy-aware uncertainty, while retaining unclustered t and KS tests only as descriptive sensitivity calculations.
How it was done	Computes H_RS separately for Subject and PCS, forms the paired difference, and compares Human-all with Baseline. Primary resampling preserves Human contributor-to-session clustering and Baseline sessions.
Result shown	Human paired mean = +0.00113; Baseline = -0.00174; contrast = +0.00287 with a hierarchy-preserving interval approximately [-0.00001,+0.00575], narrowly including zero. Unclustered Welch t=1.94, p=.0521; KS D=.0226, p=.2884. The paper treats the full-cohort result as unresolved.
Population / data	All Human versus Baseline
Source role	Primary result source

Reader note

The current paper is more explicit than the prior guide that block-level t/KS calculations do not support participant-level inference. For population interpretation, the hierarchy-preserving interval is the principal uncertainty statement.

3.6.2 / Table 4 / Figure 4

Session-count pattern and Baseline offset interpretation

Notebook and header	NB2 - Session-Count Pattern (Descriptive) - Mean Paired ΔH_RS by Condition; NB1 - Analysis 3 - Baseline Offset - Synthetic-Pipeline Check and Bootstrap Seed Stability; Exp4 vs Exp5 - Statistic 1 - Within-Baseline paired delta
What it does	Describes mutually exclusive Human session-count strata and shows how the realized Baseline contributes to the Human 5+ contrast; then checks whether a stable negative Baseline offset appears to be generated by the pipeline or recurs in a second dataset snapshot.
How it was done	NB2 calculates block-weighted ΔH_RS within Baseline, AI, Human-all, and non-overlapping Human 1-session, 2-4-session, and 5+-session groups. NB1 sends Bernoulli(.5) synthetic calls through the pipeline. The cross-dataset notebook applies the same ΔH_RS definition to an Exp5-prescreen snapshot.
Result shown	Table 4: Baseline -0.00174; AI +0.00094; Human-all +0.00113; Human 1 session -0.00004; 2-4 sessions +0.00322; 5+ sessions +0.00435. Human 5+ minus Baseline = +0.00609; the realized Baseline contributes about 29% of that contrast. Pilot 4 Baseline was -0.001741 versus -0.000459 in the Exp5-prescreen snapshot; both intervals included zero, and the synthetic-pipeline check did not show a consistent offset.
Population / data	Pilot 4 Baseline, AI, pooled Human, mutually exclusive Human strata; secondary Exp5-prescreen Baseline snapshot
Source role	NB2/NB1 primary; Exp4-vs-Exp5 secondary sensitivity source

3.6.3 / S3.1 / Table S1 / Figure S1

Shuffle sensitivity of the H_RS score

Notebook and header	NB2 - Single-Scale Hurst Delta - Shuffle Test
What it does	Changes within-block order while preserving bit composition and paired structure, asking whether the Human 5+-Baseline separation depends on ordering rather than composition alone.
How it was done	Applies the same positional permutation to Subject and PCS within nonoverlapping windows W=3,5,10,25,50,150. Each nontrivial window is repeated 200 times and the Human 5+-Baseline KS distance is recalculated.
Result shown	Unshuffled KS distance = .0541. Mean shuffled distances were .0491, .0600, .0628, .0586, .0395, and .0333 from W=3 through W=150. Separation changed with ordering and was smallest after full-block shuffling; the response was not monotonic.
Population / data	Human 5+ and Baseline blocks; 200 realizations per nontrivial window
Source role	Primary result source

3.6.4 / S4

Direct-versus-paired interpretation, stream decomposition, thresholds, and label swap

Notebook and header	NB2 - Single-Scale Hurst - 5+ Sessions; Authoritative Four-Estimand Component Table; Partial Selection-Aware Null for the Documented Session Thresholds; Label Swap Test (Observer-Dependent Invariance)
What it does	Combines Direct and Paired views of the same Human 5+ data, decomposes the paired contrast into Subject and PCS components, and calibrates selection among the documented nested 2+/3+/5+ thresholds.
How it was done	Compares Human 5+ with Baseline in direct Subject, paired Subject-minus-PCS, and separate component views. Composition-conditional order randomization preserves each stream's bit composition while assessing the fixed 5+ result and the maximum positive contrast across 2+/3+/5+. Label-swap randomizes stream labels within blocks as a sensitivity check.
Result shown	Direct KS D=.0575, p=.0243; paired KS D=.0541, p=.0398 are overlapping block-level descriptions, not independent confirmations. Hierarchy-preserving mean paired contrast = +.006090, descriptive 95% CI [+0.000776,+0.016003]. Subject was +.00315 above Baseline and PCS -0.00294 below. Fixed 5+ one-sided random-order tail area=.00695; maximum over 2+/3+/5+=.00800. This calibration does not adjust the wider exploratory program.
Population / data	Human session-count groups, Human 5+, AI, and Baseline
Source role	Primary exploratory result and selection-sensitivity source

3.6.5-3.6.8 / S3.2-S3.4

Run structure, composition, entropy, variance, and metric cross-checks

Notebook and header	NB2 - Run-Structure Concordance and Paired Robustness Check; Rule Out - Entropy; Focal paired-delta variance comparison; Alternative Ordering Metrics - Direct and Relational Three-Condition Comparison; DFA Cross-Check
What it does	Asks whether simpler run/composition summaries, excess variance, or alternative ordering estimators reproduce the exploratory Human 5+ H_RS contrast.
How it was done	Computes run count, maximum run length, run-length entropy, bit entropy/n-gram summaries, paired H_RS variance, lag-1 autocorrelation, and DFA. Hierarchy-aware resampling is used for the Human 5+-Baseline paired contrasts where applicable.
Result shown	Run-count, maximum-run, run-entropy, bit-composition, n-gram, and lag-1 measures did not show a resolved Human 5+-Baseline paired contrast. Paired H_RS variance was .004091 Human 5+ versus .004116 Baseline, ratio .994 [.865,1.172]. DFA had the same direction but a smaller standardized difference (d=.065 versus .095 for H_RS; unadjusted p=.0874). The paper therefore treats metric choice as part of the uncertainty, not as independent replication.
Population / data	Human 5+ versus Baseline, with broader condition summaries where shown
Source role	Primary characterization/sensitivity source

3.7.1 / S1

Position, weighting, recurrence, PI influence, and session order

Notebook and header	NB2 - Within-Session Position Sensitivity Analysis; Within-Session Hurst-Delta Drift - Human 5+ vs Baseline Interaction Test; Hierarchy-Preserving Bootstrap Sensitivity; Human 5+ Session Recurrence and PI-Influence Check; First Session Versus Later Sessions Among Eventual Human 5+ Contributors
What it does	Tests whether the Human 5+ pattern depends on block position within sessions, weighting choice, one session, the principal investigator, or first-versus-later session ordering.
How it was done	Uses continuous position slopes, a post-hoc early/late split, contributor-then-session bootstrap, block/equal-session/equal-contributor weighting, leave-one-session-out analyses, PI exclusion, and within-contributor first-versus-later comparisons.
Result shown	Continuous slopes were unresolved. Human 5+ minus Baseline was +.009921 early and +.002259 late; the direct late-minus-early interaction was -.007662 [-.019489,+.004125], unresolved. Block/equal-session contrast = +.00609; equal-contributor/equal-session = +.00818. Leave-one-session-out subgroup means stayed positive; excluding the PI increased the Baseline contrast to +.010423. First-versus-later trajectories were mixed.
Population / data	Three Human 5+ contributors / 28 sessions; Baseline 103 sessions; broader complete-session checks as shown
Source role	Primary sensitivity source

3.7.2 / S2.2

Lagged Subject-PCS coupling in Pilot 4

Notebook and header	NB2 - Exploratory Relational Analysis - Lagged Cross-Correlation Family; Hierarchy-Preserving Family Check - Lagged Correlation Variance
What it does	Calculates session-level Subject(t)-PCS(t+k) correlations at lags -3 through +3 and separately compares their means and between-session variances. Seven mean plus seven variance comparisons form a 14-cell family.
How it was done	Correlations never cross session boundaries. Initial Levene/mean tests are followed by hierarchy-preserving resampling of Human contributors and sessions, with Baseline sessions as whole units. Max-statistic and conservative 14-cell corrections are reported.
Result shown	At lag +2, Human variance was lower: variance ratio=.5505, Levene p=.001096, clustered interval [.383,.808]. Hierarchy-preserving p=.00170; seven-lag max-statistic p=.00780; conservative 14-cell p=.01560. The mean difference (+.0354, p=.154) was unresolved. No lag was predicted in advance, so the result remains exploratory.
Population / data	All-Human: 158 sessions from 121 contributors; Baseline: 103 sessions
Source role	Primary Pilot 4 relational source

3.7.2 / S2.1 / 4.6

Local-coupling variability across window sizes - source boundary

Notebook and header	NB2 - Exploratory Relational Analysis - Local Coupling Stability (Pilot 4 short-window analysis). Separate 80-block local-coupling dataset: not contained in the four supplied notebooks.
What it does	Distinguishes what Pilot 4 itself shows from what the current paper attributes to a separately collected longer-session dataset. This boundary changed materially from the prior reader guide.
How it was done	Pilot 4 computes nonoverlapping-window Subject-PCS correlations inside 30-block sessions. The paper supplement states that full-Human W=5 and W=10 comparisons with Baseline were unresolved. A separate 80-block dataset extends the curve to W=15 and W=20 and is the source of the reported lower Human variability at longer windows.
Result shown	For Pilot 4, the full-Human W=5 and W=10 comparisons with Baseline were unresolved. Any narrowing at those short windows should therefore be read as descriptive rather than a resolved Pilot 4 effect. The paper Supplement attributes lower full-Human variability at the longer W=15 and W=20 windows to the separate 80-block dataset; that external dataset is also where the lag-specific +2 pattern did not reproduce.
Population / data	Pilot 4 complete sessions plus a separate 80-block dataset outside the supplied notebook set
Source role	NB2 primary for Pilot 4 short-window analyses; external source required for W=15/W=20 and cross-dataset relational claims. Reconciliation note: main-text wording such as lower variability "across multiple window sizes" should be read together with S2.1, which states that Pilot 4 full-Human W=5/W=10 comparisons were unresolved.

2.11 / 3.8
Preregistered held-out synthetic diagnostic-recovery validation

Notebook and header	PDP_Diagnostic_Validation_Prereg.ipynb, in this repository's preregistration/ folder - not one of the four primary analysis notebooks included in this repository.
What it does	Tests whether the already-frozen PDP diagnostic rules can distinguish constructed marginal versus zero-lag relational changes, localize temporal relationships at lags -2 through +2, and detect programmed coupling instability.
How it was done	Validation datasets were generated after rules were frozen; classifications were made from blinded feature tables before the generating-condition key was revealed. Fixed thresholds were used for paired-H_RS shift, zero-lag correlation, lag localization, and 10-block local-correlation variance.
Result shown	Paper §3.8 reports 300/300 correct marginal-only classifications; 300/300 correct zero-lag relational-only; 1,500/1,500 correct lag localizations (300 at each of five lags); 100/100 correct coupling-instability; and 100 null controls with no marginal, zero-lag relational, or instability false-positive flags. Across 2,300 held-out datasets, every applicable preregistered criterion was met. No no-lag absence rule was claimed because none was preregistered.
Population / data	2,300 held-out synthetic datasets built from the Study 1 Baseline cohort under controlled generating conditions
Source role	Paper-supported result. The validation notebook is included in preregistration/ rather than analysis/, since it is a preregistered check locked separately from the ordinary exploratory notebooks.

Why this page is in the guide

The reader guide is meant to prevent a paper claim from being silently assigned to the wrong notebook. The new §3.8 result is prominent in the current paper and conclusion, so an explicit “not in supplied set” entry is more useful than omitting it.

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4.4

Planned preregistered replication - intended Study 2 analysis roles

Notebook and header	No Study 2 execution notebook is contained in the four supplied Pilot 4 notebooks. The Pilot 4 notebooks provide the exploratory rationale and calibration that motivated the prospective plan; the future preregistered Study 2 analysis should be checked against its own preregistration and execution materials.
What it does	Separates the planned follow-up into analysis roles. The primary confirmatory estimand is a direct, two-tailed Human-versus-Baseline comparison of Subject-stream single-scale H_RS. PCS and paired-differential H_RS comparisons are mandatory diagnostics that show how any direct difference is distributed across the two same-call streams. Relational analyses remain exploratory.
How it is intended to be done	Run the direct Subject-stream Human-versus-Baseline H_RS comparison as the primary confirmatory test without prespecifying persistence versus antipersistence direction. Examine PCS and Subject-minus-PCS H_RS as mandatory component diagnostics using the same underlying acquisitions. Separately examine window-to-window variability of local Subject-PCS correlations across prespecified window sizes and temporal-lag relationships across a prespecified lag range as exploratory relational analyses.
Interpretive rule	Do not treat Direct, PCS, and paired-differential results as independent confirmations because they use overlapping calls. The relational analyses ask a different joint-stream question but remain exploratory in the current research program. Pilot 4 subgroup, threshold, lag +2, and wider exploratory-battery results are rationale/calibration, not the Study 2 confirmatory battery.
Population / data	Prospective new Study 2 Human and contemporaneous Baseline data, with longer fixed-length sessions, repeated participation, and inference that preserves blocks within sessions and sessions within participants, as specified in the current paper.
Source role	Paper-defined prospective analysis plan. The supplied Pilot 4 notebooks are supporting rationale/calibration only; they are not the execution source for the future confirmatory result.

4.7 and 5.11

Interpretation and limitation of contiguous stream partitioning

Notebook and header	Interleaving - 1. Reconstruct original and PRNG-interleaved streams; 2. Original vs interleaved Baseline summary; 6. Reconstruction-null test; Deterministic odd/even check; 7. AI condition; 8. Baseline stream-length variance scan
What it does	Reconstructs Pilot 4 raw calls under the original contiguous split and locally balanced alternatives to test how a retrospective early/late Baseline pattern changes when positions are redistributed.
How it was done	Rebuilds retained Baseline calls under locally balanced pseudorandom partitions, repeats the reconstruction-null comparison 500 times, runs deterministic odd/even allocation, repeats the position check in AI, and scans H_RS variability across stream lengths.
Result shown	Original Baseline early-minus-late = -.004647. One interleaved reconstruction = +.001362; across 500 reconstructions median=-.000163 with 95% range [-.004166,+.004026], empirical two-sided tail area=.03. Deterministic odd/even = +.001418. The pattern was attenuated, but the paper explicitly treats this as retrospective and does not infer that interleaving is generally superior or that contiguous partitioning caused the original pattern.
Population / data	Pilot 4 Baseline raw calls; AI for the condition check
Source role	Primary source for interleaving discussion and contiguous-partition limitation

5.10

Finite-sample behavior of the single-scale H_RS score

Notebook and header	NB2 - Bounded Differential-Response Calibration of the R/S Ordering Score (primary source for the quoted §5.10 result); NB1 - Latent-H Difference and Estimator Calibration (related generator-specific calibration, not the direct source of the quoted §5.10 numbers)
What it does	Tests the exact 150-bit implemented score under bounded known generators and asks whether reversing which stream receives stronger ordering structure creates estimator asymmetry large enough to explain the Human 5+ paired difference.
How it was done	NB2 uses rank-binarized Gaussian AR(1) and fractional-Gaussian-noise sequences while reproducing observed Human 5+ composition margins; it compares forward and reverse parameter assignments. NB1 contains a related thresholded-fGn latent-parameter calibration with an explicit interpretation boundary.
Result shown	The score followed imposed ordering contrasts in the expected direction. Every directional-asymmetry interval included zero; the largest endpoint was about .00152 versus the observed Human 5+ paired difference +.00435. Within the simulated cases, estimator asymmetry did not account for the full observed contrast. The paper also notes that the two persistence dose .35 injection cells remain precision-limited by the 103 real Baseline sessions.
Population / data	Synthetic 150-bit sequences; composition margins calibrated to Human 5+ where specified
Source role	NB2 primary source for the paper's bounded differential-response and directional-asymmetry claim; NB1 is a related calibration source but should not be used to reproduce the specific §5.10 AR(1)/fGn composition-matched numbers.

Cross-dataset sensitivity context

Pilot 4 versus Exp5-prescreen Baseline comparison

Notebook and header	Exp4 vs Exp5 - Statistic 1 - Within-Baseline paired delta; Statistic 2 - Baseline level vs Human PCS level
What it does	Reconstructs Pilot 4 Baseline from frozen raw calls and compares it with a dated, ongoing Exp5-prescreen snapshot. It checks whether the negative realized Pilot 4 Baseline delta looks like a stable processing offset and whether Baseline levels differ from Human PCS levels.
How it was done	Processes both datasets with the same paired H_RS definition; bootstraps sessions; compares within-Baseline paired means and Baseline-versus-Human-PCS levels under the available contributor definitions.
Result shown	Within-Baseline paired means: Pilot 4 -0.001741, 95% CI [-0.003811,+0.000355]; Exp5-prescreen -0.000459 [-0.002286,+0.001408]. Baseline-minus-Human-PCS intervals included zero in both datasets and under the reported Exp5 deduplication variants.
Population / data	Pilot 4 Baseline and dated Exp5-prescreen snapshot
Source role	Secondary cross-dataset sensitivity source; not primary Pilot 4 inference

S3.4 / 3.6.8

Alternative ordering-estimator context

Notebook and header	NB2 - Alternative Ordering Metrics - Direct and Relational Three-Condition Comparison; DFA Cross-Check - Does the Human5+ Pattern Replicate Under a Bias-Flatter Estimator?
What it does	Checks whether the Human 5+ ordering pattern is specific to the single-scale H_RS implementation or appears in simpler/alternative ordering measures.
How it was done	Computes lag-1 autocorrelation and entropy-family metrics in Direct and Paired forms, then evaluates DFA separately as an estimator cross-check. These analyses are exploratory and share the same underlying data.
Result shown	Lag-1 autocorrelation, n-gram entropy, and entropy-rate slope did not resolve the Human 5+-Baseline contrast. DFA pointed in the same direction but with smaller standardized separation (d=.065 versus d=.095 for H_RS) and unadjusted p=.0874. The current paper uses this to support a metric-dependence caution, not a replication claim.
Population / data	Human 5+ and Baseline, with three-condition summaries for several alternative metrics
Source role	Primary sensitivity source

Known navigation and reconciliation notes

Items a reader should not silently merge

Separate 80-block relational dataset

The current paper's S2.1 longer-window local-coupling statements (W=15 and W=20) concern a dataset outside Pilot 4. In Pilot 4 itself, full-Human W=5 and W=10 comparisons were unresolved. The paper also says the lag-specific +2 pattern did not reproduce in the separate dataset. Those claims are not expected in NB1, NB2, or the Interleaving notebook.

Held-out synthetic diagnostic-recovery validation

Paper §§2.11 and 3.8 report a preregistered blinded validation totaling 2,300 held-out synthetic datasets. This analysis is not in NB1, NB2, or the Interleaving notebook - it is preregistration/PDP_Diagnostic_Validation_Prereg.ipynb in this repository. Look there, rather than in NB1/NB2, for the §3.8 counts.

Artifact-injection preregistration

Unlike the prior guide, the current paper includes the locked 32-cell result: 30 PASS, 2 INDETERMINATE, 0 FAIL. NB1 contains the preregistered grid and its complete both-metric breakdown.

Same-position mutual information

The current paper no longer summarizes the entire MI battery as simply unresolved. Human excess same-position MI is individually negative/resolved, but the direct Human-Baseline and AI-Baseline contrasts are unresolved. The between-condition conclusion is therefore still null at the tested resolution.

Interleaving notebook

This remains the direct source for attenuation under locally balanced interleaving, 500 reconstruction-null repetitions, deterministic odd-even allocation, the AI check, and the contiguous-partition limitation. It is retrospective and does not establish causation or general superiority.

Outdated notebook cross-reference

NB1 still contains a visible header "Empirical physical-half calibration for Paper §3.3.4." In the current paper, the empirical physical-half result is in §3.2.2. Search by the header text, but use the current paper section number when citing the manuscript.

Cross-dataset notebook

The Exp5-prescreen values are secondary sensitivity context for Baseline-offset interpretation. They are not part of the Pilot 4 primary inferential dataset and should not be used to upgrade a Pilot 4 result.

Study 2 analysis-role boundary

The current paper defines one primary confirmatory follow-up test: the direct, two-tailed Human-versus-Baseline Subject-stream H_RS comparison. PCS and paired-differential analyses are mandatory diagnostics, not independent confirmations, and relational window/lag analyses remain exploratory. Do not infer a future confirmatory battery from the breadth of the Pilot 4 exploratory notebooks.